Homework 5

October 21, 2026

Problem 1

Let

$$B(\lambda) = \begin{pmatrix} \lambda^3 + \lambda & \lambda^2 \\ \lambda^2 - 3\lambda & 2\lambda \end{pmatrix}, \qquad C(\lambda) = \begin{pmatrix} 1-\lambda & \lambda^2 & \lambda \\ 2\lambda - 1 & \lambda(1-\lambda) & -2\lambda \\ 1+\lambda^2 & \lambda^2 & -\lambda^2 \end{pmatrix},$$

and

$$A(\lambda) = \begin{pmatrix} B(\lambda) & 0 \\ 0 & C(\lambda) \end{pmatrix}.$$

Find out

(1) the Smith canonical form of $B(\lambda)$;

(2) the elementary divisors of $A(\lambda)$.

Problem 2

Let $D : C[\lambda] \to C[\lambda]$ be defined by

$$D\Big(\sum_{i=0}^{n} \alpha_i \lambda^i\Big) = \sum_{i=1}^{n} i\,\alpha_i \lambda^{i-1}.$$

Show that D is a linear transformation on $C[\lambda]$, and determine $\ker[D]$.

Problem 3

Let $\mathscr{A} \in L(C_n[\lambda])$ be defined, for a nonzero constant $a \in C$, as

$$\mathscr{A}[f(\lambda)] = f(\lambda + a), \qquad \forall\, f(\lambda) \in C_n[\lambda].$$

Find out

(1) the matrix representation of $\mathscr{A}$ with respect to the basis $1, \lambda, \lambda^2, \ldots, \lambda^n$;

(2) the eigenvalues and the associated eigenvectors of $\mathscr{A}$.

Problem 4

Let $V = C_n[s]$ and $D : C_n[s] \to C_n[s]$ be defined by

$$D[f(s)] = n a_n s^{n-1} + (n-1) a_{n-1} s^{n-2} + \cdots + 2a_2 s + a_1,$$
$$f(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_2 s^2 + a_1 s + a_0.$$

Find out

(1) the elementary divisors of D;

(2) the eigenvalues and the associated eigenvectors of D.